
MORGAN PFAFF

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EDUCATION

University of California (UC) Berkeley and UC San Francisco

Berkeley, CA

Doctor of Philosophy in Bioengineering

Expected October 2026

- Dissertation: “Viscoelastic, Semi-synthetic HyA Hydrogels for Soft Tissue Regeneration”
- Advisor: Kevin E. Healy, PhD
- GPA 4.0/4.0
- National Science Foundation Graduate Research Fellowship Awardee
- Major area of study: Biomaterials Engineering and Tissue Regeneration
- Minor area of study: Cell, Tissue, and Developmental Biology

Northeastern University

Boston, MA

Bachelor of Science in Biochemistry

May 2021

- GPA 4.0/4.0
- Honors Program; Honors in the Discipline
- Engineering study abroad program, Oxford, United Kingdom (Summer 2019)

RESEARCH EXPERIENCE

Kevin Healy Lab, UC Berkeley

Berkeley, CA

PhD Candidate

January 2022 – August 2026

- Engineered and mechanically characterized tunable, viscoelastic, semi-synthetic hyaluronic acid (HyA) hydrogels to promote muscle regeneration and vascularization after delayed rotator cuff repair and volumetric muscle loss, including formoterol-loaded formulations evaluated for drug release, mechanics, and injectability
- Investigated hydrogel-driven mechanisms of muscle regeneration, including brown adipose tissue (BAT) differentiation of fibro-adipogenic progenitors (FAPs), in 3D in vitro culture; performed all single-cell RNA-seq analysis of patient-derived FAPs and used the findings to guide hydrogel design
- Prepared hydrogels for, and analyzed and interpreted data from, in vivo injury studies conducted with orthopedic surgery collaborators
- Published 2 peer-reviewed papers, gave 5 presentations at 4 national and international conferences (BSRM, BMES, SFB, ORS), and co-invented the hydrogel technology now in development at MuscleMatrix
- Mentored 3 rotation students, 1 undergraduate, 1 visiting international PhD student, and a Master of Translational Medicine (MTM) team, and trained dozens of other students on equipment and methods

Randall Erb Lab, Northeastern University

Boston, MA

Biomaterials Undergraduate Researcher

September 2016 – July 2021

- Developed a fabrication protocol for dual-network hydrogels reinforced with magnetically responsive hydroxyapatite fibers for bone graft applications, and characterized their mechanics using dynamic mechanical analysis and scanning electron microscopy
- Designed and analyzed COMSOL finite element simulations of hydroxyapatite fiber rotation in an alginate matrix to investigate a potential new toughening mechanism
- Fabricated freeze-cast conduits with highly aligned, double-layered porosity for peripheral nerve repair in collaboration with Dartmouth College and Harvard Medical School
- Secured independent research funding by writing a successful proposal for the competitive Project-Based Exploration for the Advancement of Knowledge (PEAK) Summit Award

Marc Hansen Lab, University of Connecticut Health Center

Farmington, CT

Summer Research Fellow

Summers 2017 & 2018

- Piloted synthetic-lethal experiments testing inhibitor and siRNA combinations in osteosarcoma cell lines and hFOB cells to identify druggable vulnerabilities in growth factor receptor signaling, using PCR and absorbance assays
- Used laser capture microdissection on histological sections to isolate osteoblast and osteoclast populations and investigate cell behavior and genetics in Paget's disease of bone
- Presented results at the program's poster session; work provided preliminary data for the lab's NIH grant applications

TEACHING EXPERIENCE

University of California, Berkeley

Berkeley, CA

Reader

Fall 2026

- BIOE/MSE C118/C208: Biological Performance of Materials (Instructor: Dr. Kevin Healy; 33 students): grade homework, exams, and team design projects, including written reports and presentations
- BIOE 252: Clinical Need-Based Therapy Solutions (Instructor: Dr. Syed Hossainy; ~100 students): form and guide student teams developing therapy concepts for unmet clinical needs, grade homework and team projects, and answer student questions

Guest Lecturer

2023 – 2026

- Delivered 4 full-length lectures in BIOE/MSE C118/C208 on hydrogels (2), polymers and viscoelasticity, and an introduction to polymers, in place of the course instructor, Dr. Kevin Healy

Graduate Student Instructor

Fall 2022

- BIOE/MSE C118/C208: Biological Performance of Materials (Instructor: Dr. Kevin Healy)
- Led a weekly discussion section reviewing lecture material and held weekly office hours for a class of ~40 students
- Administered and graded homework and exams, and evaluated ~10 team design projects, including 15-page reports proposing biomaterial solutions to unmet clinical needs and a series of pitch, 10-minute, and 20-minute presentations

TRANSLATIONAL, REGULATORY & INDUSTRY EXPERIENCE

MuscleMatrix

Scientific Consultant

August 2026 – Present

- Consult for an early-stage startup spun out of my doctoral research that is developing an injectable, cell-free hyaluronic acid scaffold to support muscle repair after rotator cuff injury
- Help build company operations from the ground up, including regulatory compliance, environmental health and safety (EHS), laboratory buildout and equipment procurement, and manufacturer and supplier sourcing
- Contribute to FDA regulatory strategy for approval as a medical device or combination product, and write grant applications to secure funding
- Conduct formulation development, characterizing hydrogel injectability and mechanical properties and evaluating how drug loading alters material properties to inform a planned drug-loaded version
- Onboard and train new scientific staff on the hydrogel platform and laboratory operations

Lake Street Labs

Scientific Consultant

June 2026 – August 2026

- Conducted a Phase 1 scientific and regulatory feasibility review of a novel oral health product for an early-stage startup, delivering a written feasibility report with recommendations
- Reviewed clinical and preclinical literature and evaluated candidate formulations for chemical compatibility, stability, and shelf-life testing needs
- Assessed FDA regulatory-pathway considerations, including product classification and permissible claims

Vertex Pharmaceuticals

Boston, MA

Exploratory Research Co-op

January 2020 – June 2020

- Developed and optimized cellular assays to detect transcription factors in potentially druggable pathways for sickle cell disease (SCD), and profiled protein expression over time in pilot experiments
- Generated stable cell lines by molecular cloning of recombinant vectors for genomic integration
- Independently planned, executed, and analyzed experiments, and presented results and journal club articles to the SCD exploratory research team

Editas Medicine

Cambridge, MA

Screening Group Co-op

January 2018 – June 2018

- Screened sgRNAs to identify the most effective candidates for the company's pipeline diseases, introducing CRISPR-mediated edits via nucleofection, transfection, and viral transduction
- Programmed and optimized high-throughput screening assays on automated liquid handlers, including a comparative performance evaluation of the Biomek i7 and its Biomek FX predecessor
- Analyzed data in Prism, Tableau, and FlowJo and presented findings to the screening group and other departments

PUBLICATIONS & PRESENTATIONS

Publications

1. Wang, V. Y., Zhu, Y., Spaulding, K. A., **Pfaff, M. R.**, Neiman, G., Takaba, K., Chen, N., Parekh, K., Hasaballa, A., Haraldsson, H., Baker, A., Saloner, D., Wallace, A. W., Ziats, N. P., Healy, K. E., Ratcliffe, M. B. NIPAAm-co-PEG4000 thermoresponsive hydrogel with ROS scavenging properties improves in vivo border zone contractility and reduces myocardial remodeling in sheep after MI. *Acta Biomaterialia*, 2026, 211, 48–60. doi.org/10.1016/j.actbio.2025.06.046
2. **Pfaff, M. R.**, Wague, A., Davies, M., Killaars, A. R., Ning, D., Garcia, S., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Viscoelastic HyA hydrogel promotes recovery of muscle quality and vascularization in a murine model of delayed rotator cuff repair. *Advanced Healthcare Materials*, 2025, 14, 2403962. doi.org/10.1002/adhm.202403962

Oral Presentations

1. **Pfaff, M. R.**, Wague, A., Davies, M., Killaars, A. R., Ning, D., Garcia, S., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Viscoelastic HyA hydrogel promotes recovery of muscle quality and vascularization in a murine model of delayed rotator cuff repair. Oral presentation presented at: Biological Scaffolds for Regenerative Medicine (BSRM); May 2025; Napa, CA.
2. **Pfaff, M. R.**, Killaars, A. R., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Tunable semi-synthetic hyaluronic acid hydrogels promote muscle regeneration after rotator cuff injury. Oral presentation presented at: Biomedical Engineering Society (BMES) Annual Meeting; October 11–14, 2023; Seattle, WA.
3. **Pfaff, M. R.**, Killaars, A. R., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Semi-synthetic hyaluronic acid hydrogels promote muscle regeneration by inducing BAT differentiation of FAPs. Oral presentation presented at: Society For Biomaterials Annual Meeting & Exposition, "Biomaterials for Regenerative Engineering"; April 21, 2023; San Diego, CA.
4. **Pfaff, M. R.**, Faust, J. L., Erb, R. M. Dual-network hydrogels reinforced with field-responsive hydroxyapatite exhibiting high strength and stiffness while fully hydrated. Oral presentation presented at: Materials Research Society 2020 Spring/Fall Meeting and Exhibit; November 28, 2020; Virtual.
5. **Pfaff, M. R.**, Faust, J. L., Erb, R. M. Dual-network hydrogels reinforced with field-responsive hydroxyapatite exhibiting high strength and stiffness while fully hydrated. Oral presentation presented at: Northeastern Regional Student Conference of the Society for Experimental Mechanics; June 22, 2020; Dartmouth, MA.

Poster Presentations

1. **Pfaff, M. R.**, Killaars, A. R., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Tunable semi-synthetic hyaluronic acid hydrogels promote muscle regeneration after rotator cuff injury. Poster session presented at: ORS 50th International Musculoskeletal Biology Workshop; July 22–27, 2023; Midway, UT.

2. **Pfaff, M. R.**, Killaars, A. R., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Semi-synthetic hyaluronic acid hydrogels promote muscle regeneration by inducing BAT differentiation of FAPs. Poster session presented at: Society For Biomaterials (SFB) Annual Meeting & Exposition; April 2023; San Diego, CA.
3. **Pfaff, M. R.**, Erb, R. M. Understanding rod rotation in fiber reinforced hydrogels: a potential toughening mechanism. Poster session presented at: Research, Innovation, and Scholarship Expo; April 15, 2021; Virtual.
4. **Pfaff, M. R.**, Alander, C. B., Hansen, M. F. Synthetic lethal approaches to treating osteosarcoma. Poster session presented at: University of Connecticut Health Center Summer Research Fellowship Program; July 27, 2017; Farmington, CT.
5. **Pfaff, M. R.**, Faust, J. L., Erb, R. M. Dual-network hydrogels with functionalized CaP rods for bone graft applications. Poster session presented at: Research, Innovation, and Scholarship Expo; April 13, 2017; Boston, MA.
6. **Pfaff, M. R.**, Faust, J. L., Erb, R. M. Dual-network hydrogels with functionalized CaP rods for bone graft applications. Poster session presented at: National Conference on Undergraduate Research; April 6–8, 2017; Memphis, TN.

FELLOWSHIPS & AWARDS

National Science Foundation Graduate Research Fellowship (\$138,000)	2021
Research, Innovation, and Scholarship Expo (RISE) Excellence in Research Award (\$2,000)	2021
PEAK Summit Award, undergraduate research grant (\$3,000)	2020
PEAK “Shout It Out!” Award, conference travel grant	2020
RISE Innovation Alley Selection (poster selected for highest impact and commercial potential)	2017

LEADERSHIP & SERVICE

Sigma Xi, The Scientific Research Honor Society (Associate Member) **March 2020 – Present**

Vice President, UC Berkeley Chapter 2021 – Present

- Co-organize the chapter’s semi-annual Grants in Aid of Research (GIAR) program, which awards research grants of up to \$750 to undergraduate, master’s, and PhD students across STEM
- Founded and chair the Committee for Student Advancement (CSA), growing it to as many as 40 students, including preliminary members not yet eligible for associate membership; host networking events and student and guest talks, including a talk by the Editor-in-Chief of *American Scientist*
- Independently planned and executed the Sigma Xi–ATP Bio Research Symposium & Poster Session, an event open to students at all levels, from securing a funding partner to recruiting presenters and coordinating day-of logistics; drew ~25 presenters, a keynote speaker, and ~150 attendees who voted for a best-poster prize
- Led chapter recruitment through outreach to campus organizations and departments, and manage chapter operations, including meeting scheduling, event logistics, and records

Diversity & Inclusion Chair, Northeastern University Chapter 2020 – 2021

- Expanded outreach to first-generation, culture-specific, and other underrepresented student organizations, and hosted a campus-wide diversity, equity, and inclusion panel

Food Not Bombs **September 2024 – Present**

- Lead a weekly serving shift for Food Not Bombs East Bay, picking up meals prepared by a team of volunteer cooks and serving 25–40 free vegetarian meals per shift to unhoused and low-income community members
- Transformed an unstaffed curbside food drop-off into an in-person served meal, bringing a table and serving each meal directly to create a more personal experience and build ongoing relationships with community members

Bay Area Bioscience Education Health & Science Challenge (BioEHSC) **April 2022 – April 2026**

- Academic Judge: evaluated 10–15 high school team pitches each spring in a full-day bioscience pitch competition, assessing depth of knowledge through Q&A and providing feedback

Northeastern University Housing and Residential Life

Boston, MA

Resident Assistant

August 2017 – July 2019

- Served as a live-in mentor and resident advisor for ~45 freshman women a year from around the world pursuing careers in STEM, maintaining a safe, orderly environment and confronting and resolving issues surrounding diversity, racism, substance abuse, academic performance, and social acceptance
- Designed and implemented educational, recreational, and networking programs, including Q&A panels with graduate students and professors, leadership and team-building workshops, and faculty mentorship events
- Helped residents set and achieve short- and long-term goals, connect with established scientists and professors, and access resources to pursue their academic and scientific aspirations

Adventure Challenge Volunteer

September 2018 – December 2019

- Planned and led weekly swimming and recreational activities with a teen with autism, promoting physical activity, social interaction, and overall wellness

Advancement Via Individual Determination (AVID) Tutor

September 2018 – December 2018

- Led a weekly group of 6–8 students at Fenway High School in the AVID college-readiness program, teaching collaborative problem-solving and critical thinking for math and science

Northeastern University Club Sports

September 2016 – June 2021

- Triathlon Team: President/Captain (2019–2020), Chief Financial Officer (2018–2019), and Chief Procurement Officer (2017–2018)
- Club Sports Council (2019–2021): served as liaison between club sports teams and Club Sports administrators and directors

CLINICAL EXPERIENCE

Tufts Medical Center, Oncology and Hematology

Boston, MA

Clinical Care Technician

January 2019 – December 2019

- Provided direct care for 6–10 oncology and hematology patients per shift, including support with mobility, hygiene, and meals
- Monitored vital signs and blood glucose, escalated abnormal findings to nursing staff, and supported clinical teams during examinations, procedures, and emergencies

Fallon Ambulance Service

Quincy, MA

Emergency Medical Technician Basic

September 2018 – April 2019

- Assessed, treated, and transported ill and injured patients, providing immediate medical and psychological care en route to hospitals and specialty facilities
- Documented vitals, symptoms, medical history, allergies, and medications per company protocols while maintaining a calm, safe environment for patients and families

Dental Office of Dr. Andrew J. Ponichtera DMD, MS

Simsbury, CT

Dental Assistant

June 2016 – August 2016

- Supported chairside procedures including fillings, root canals, and placement of crowns, implants, bridges, and dentures
- Managed front-desk operations and supply inventory, including patient scheduling, records, and ordering

SKILLS

Biomaterials Synthesis & Mechanical Testing

- Hydrogel synthesis and chemical modification, drug loading and release, rheology, injectability testing, mechanical testing (dynamic mechanical analysis, tensile, burst, and lap shear), freeze casting, lyophilization

Materials Characterization & Microfabrication

- NMR spectroscopy, X-ray photoelectron spectroscopy (XPS), scanning electron microscopy (SEM), atomic force microscopy (AFM), microfabrication and photolithography

Cell & Molecular Biology

- Primary and 3D cell culture, immunocytochemistry, confocal and fluorescence microscopy, histology, flow cytometry, qPCR/ddPCR, Western blot, molecular cloning, CRISPR-Cas9 gene editing (transfection, nucleofection, viral transduction), automated liquid handling

Data Analysis & Computation

- R and Python (including single-cell RNA-seq analysis), GraphPad Prism, MATLAB, COMSOL Multiphysics (finite element analysis)

Regulatory & Translational

- FDA regulatory pathways and product classification, environmental health and safety (EHS) compliance, laboratory buildout and procurement, grant writing